

Assignment Title: Issue Tracker API

Goal:

Build a small backend service using Python (any web framework) and PostgreSQL. The app should let users manage issues, comments, and labels, with real-world coding challenges like concurrency, transactions, and CSV imports.

Core Features

1. Issue Management	CRUD operations on issues. Implement versioning for optimistic concurrency control.
2. Comments	Add comments to issues with non-empty bodies and author validation.
3. Labels	Unique labels that can be assigned to issues.
4. Bulk Update	Transactional bulk status update. Rollback if any issue violates rules.
5. CSV Import	Upload CSV for issue creation. Validate each row and return a summary report.
6. Reports	Endpoints for top assignees and average resolution time.

Database Requirements

Use PostgreSQL with tables: users, issues, comments, labels, issue_labels. Add unique and foreign key constraints, and indexes for performance.

Expected Endpoints

Endpoint	Method	Description
/issues	POST	Create new issue
/issues	GET	List issues (filter + pagination)
/issues/{id}	GET	Get issue with comments & labels
/issues/{id}	PATCH	Update issue (with version check)
/issues/{id}/comments	POST	Add comment
/issues/{id}/labels	PUT	Replace labels atomically
/issues/bulk-status	POST	Bulk status update (transactional)
/issues/import	POST	CSV upload for issue import
/reports/top-assignees	GET	Aggregated report
/reports/latency	GET	Average time report

Evaluation Criteria (100 pts)

Area	Points
API correctness (CRUD, filtering, CSV)	30
Concurrency & transactions	25

Code structure & clarity	20
Error handling & validation	15
Tests & documentation	10

Deadline: 48 hours after receiving the assignment.

Bonus: Add a `/issues/{id}/timeline` endpoint showing issue history.